

15 Answer: D

An oscillating system will have the same frequency as that of the periodic force applied to it. Thus the system will have a frequency of 2.50 Hz after being subjected to the force, and period

$$T = 1/2.5 = 0.40 \text{ s}$$

$$\text{So } 0.15 \text{ s} = 0.375 T$$

$$\begin{aligned} v &= 1.5 (2\pi/0.4) \cos (2\pi/T (0.375T)) \\ &= 16.7 \text{ cm s}^{-1} \end{aligned}$$

As $t = 0.15 \text{ s}$ is less than half a period, the mass would still be moving upwards.